

Ben Racine

Software engineer focused on high-assurance, safety-critical, and distributed systems.

EXPERIENCE

CIMARRON SOFTWARE SERVICES → AXIOM SPACE — Houston, TX — 3/2023-present

- Architected, designed, and implemented the command and telemetry middleware integrating the Commercial Crew UI with the Commercial network and the Critical Crew UI with the Flight Control network, enabling reliable command execution and real-time telemetry.
- Developed the Ground, Commercial Crew, and Critical Crew user interfaces around shared feature libraries, reducing maintenance costs and providing a consistent experience.
- Established automated software assurance pipelines that enforce testing, documentation, security analysis, and controlled deployment, improving traceability, release readiness, and software quality.
- Led the verification and testing strategy for flight-critical user interfaces through Critical Design Review (CDR), ensuring objective evidence of requirements verification.
- Implemented the Axiom Design Language, creating a reusable UI component library that improved interface consistency, accelerated development, and achieved 99% automated unit test coverage.
- Mentored engineers in applying type-driven design to encode safety-critical invariants, shifting correctness from runtime validation to compile-time verification.
- Developed SCADA-style operator interfaces with real-time telemetry visualization for the AxEMU Display and Control Unit (DCU) trainer, supporting crew training and mission operations.

WEATHER SOURCE — Salem, NH — 4/2021-3/2023

- Developed a mission-specific weather alerting system with team-wide notifications.
- Published type-safe geographic and meteorological APIs with OpenAPI documentation.
- Applied functional programming to ensure modular, verifiable, maintainable systems.
- Built microservices for auth, billing, mapping, data transforms, and weather-rule logic.
- Deployed a secure Snowflake–DWAS portal as the single source for client analytics.
- Scaled and hardened production to support and retain multiple Fortune 500 clients.

RACINE DESIGNS — Box Elder, SD — 4/2020-3/2021

- Developed online learning software using C#, React, Redux, and Azure for enterprise clients.
- Enforced strict TDD discipline, sustaining 80-100% coverage for React components.
- Delivered high-volume software on tight timelines, ensuring consistent, high-quality output.
- Led biweekly client demos, driving alignment and satisfaction across stakeholders.
- Founded and operated Racine Designs LLC, managing full-cycle projects and client relations.

CORNERSTONE SYSTEMS NW — Lynden, WA — 4/2011-3/2019

- Enhanced CISNET's C++ Smoking History Generator for lung cancer research modeling.
- Modeled U.S. smoking-policy outcomes using C++ and Python simulation frameworks.
- Shipped Daycast.com, a real-time planner using Node, SQL, WebSockets, and Electron.
- Delivered a single-page application that meets complex PHMSA regulatory requirements.

- Developed genetic-simulation frameworks for the National Cancer Institute
- Created a web-based data visualization dashboard for Massachusetts General Hospital cancer-surveillance analytics.

GLOSTEN — Seattle, WA — 11/2006-3/2011

- Predicted hydro-elastic loads on the SR520 floating bridge for Washington State DOT.
- Optimized anemometer performance for Boeing's Sea-Based X-Band (SBX) Radar using OpenFOAM CFD and parallel simulation on a custom Linux Beowulf cluster.
- Designed underwater transducer fairings for NOAA research vessels using CFD.
- Built a tool mapping FEA results from MAESTRO to NASTRAN for detailed fatigue analysis.
- Successfully authored proposals securing over \$85k in engineering services.

SKILLS AND TECHNOLOGIES

- Languages: Rust, C++, TypeScript, C#, Python
- Systems & Infrastructure: Linux, Docker, Kubernetes, GitLab CI/CD, AWS, GCP, Azure
- Distributed Systems & Observability: Tokio, NATS, WebSockets, SSE, OpenTelemetry, Grafana
- Data & Storage: PostgreSQL, Snowflake, SQLite, SQL Server, Cosmos DB
- Frontend & Testing: React, Vite, Vitest, Playwright, React Testing Library, Radix UI
- Engineering & Simulation: MATLAB, ANSYS, Femap, OpenFOAM, NASTRAN
- Platforms & Security: Keycloak, Stripe, Twilio

EDUCATION

MS in Computational Fluid Dynamics, Pennsylvania State University, State College, PA

BS in Naval Architecture and Marine Engineering, Webb Institute, Glen Cove, NY

REFERENCES

Mark Gibbs (Weather Source), Shawn Freeman (Weather Source), Stephen Schutt (CSNW), John Cross-Whiter (Glostten), and Dr. Eric G. Paterson (PSU-ARL)

PUBLICATIONS

CFD-Based Methods for Simulation of Marine-Vehicle Maneuvering, B. Racine and E. Paterson, AIAA 2005-4904, 35th AIAA Fluid Dynamics Conference and Exhibit, June 2005, Toronto, Canada

Application of the Free Wave Spectrum to Minimize and Control Wake Wash, Justus Heimann, Bruce L. Hutchison, and Benjamin J. Racine, Paper No. SMTc-049-2008, SNAME Annual Meeting and Expo, 15-17 October 2008, Houston, TX

Holford, T. R., Levy, D. T., McKay, L. A., Clarke, L., Racine, B., Meza, R., ... Feuer, E. J. (2014). Patterns of birth cohort-specific smoking histories, 1965-2009. American journal of preventive medicine, 46(2), e31-e37. doi:10.1016/j.amepre.2013.10.022

Peng, B., Chen, H. S., Mechanic, L. E., Racine, B., Clarke, J., Clarke, L., ... Feuer, E. J. (2013). Genetic Simulation Resources: a website for the registration and discovery of genetic data simulators. Bioinformatics (Oxford, England), 29(8), 1101-1102. doi:10.1093/bioinformatics/btt094

Peng, B., Chen, H. S., Mechanic, L. E., Racine, B., Clarke, J., Gillanders, E., & Feuer, E. J. (2015). Genetic data simulators and their applications: an overview. Genetic epidemiology, 39(1), 2-10. doi:10.1002/gepi.21876